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Printed Pages – 4 Roll No. : 3112414039

324355(24)

B. E. (Third Semester) Examination, Nov.-Dec., 2015

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL ENGG. MATERIAL

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory and is of 2 marks. Attempt any two among parts from (b), (c) and (d) each of 7 marks.

1. (a) What is meant by relaxation time? 2

(b) Give an account of the phenomenon of super-conductivity and its behaviour in magnetic filed. Give some applications of this phenomenon. 7

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(c) Explain 'mean free path' of an electron in a conducting material and deduce expression for same. Show that resistivity of a metal varies with temperature. 7

(d) (i) A uniform silver wire has resistivity of $1.4 \times 10^{-16} \Omega - \text{cm}$ at room temperature. An electric field of 50 V/m exists along the length of wire. If the number of free electrons is $5.8 \times 10^{28}/\text{m}^3$, calculate the drift velocity of electrons, their mobility relaxation time. 4

(ii) A silver wire has length 2 cm and resistance 4 ohm. The diameter of uniform circular cross-section is 0·3 mm, find the condutivity of material. 3

2. (a) What happens when a free electron meets a hole? 2

(b) Describe Hall effect. Explain how Hall effect is utilized for determining the type of semiconductor specimen (P-type or *n*-type) 7

(c) What is meant by semiconductor? What are *n*-type and *p*-type semiconductor? 7

(d) Explain diffusion in semiconductors. A semiconductor at room temperature has intrinsic carrier concentration

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of $2 \cdot 4 \times 10^{16}/m^3$. If the electron mobility in the semiconductor is $0 \cdot 21\ m^2$/volt-sec and hole mobility is $0 \cdot 045\ m^2$/volt-sec, compute the resistivity at room temperature. 7

3. (a) Define the term electric dipole moment. 2

(b) Mention the conditions for occurance of electronic polarization and show that

$$\alpha_e = \frac{\varepsilon_0 (\varepsilon_r - 1)}{N}$$

where α_e is electronic polarizibility, ε_r is relative permittivity and N is number of atoms per m^3. 7

(c) Explain dielectric losses. What is the physical significance of loss angle? 7

(d) Explain breakdown in dielectrics. A gas containing $2 \cdot 7 \times 10^{25}$ atoms per m^3 has polarizability of $2 \cdot 7 \times 10^{-40}\ F - m^2$. Calculate its relative dielectric constant. 7

4. (a) Define magnetization. 2

(b) Explain the terms diamagnetism, paramagnetics, ferro-

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magnetism with reference to magnetic dipoles of atoms. 7

(c) Explain hysteresis in ferromagnetic materials. What are soft and hard magentic materials? 7

(d) What are ferrites? A magnetic material has a magnetization of 2,300 A/m and produces a flux density of $0 \cdot 00314\ Wb/m^2$. Calculate the mangetizing force and relative permeability of the material. 7

5. (a) Give two applications of photoemissive material. 2

(b) What are photoemitters? Explain junction-diode photoemitters. 7

(c) Write short notes on : 7

(i) Electro luminescence

(ii) Junction Laser

(d) Describe construction and working of fluorescent lamp. 7

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